A framework for feature selection in clustering

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- The problem and the solution we want
- Introduce sparse clustering framework
- Sparse k-means and hierarchical clustering
- Comparison to previous work
- Experiments - good clustering ability, variable selection

The Problem

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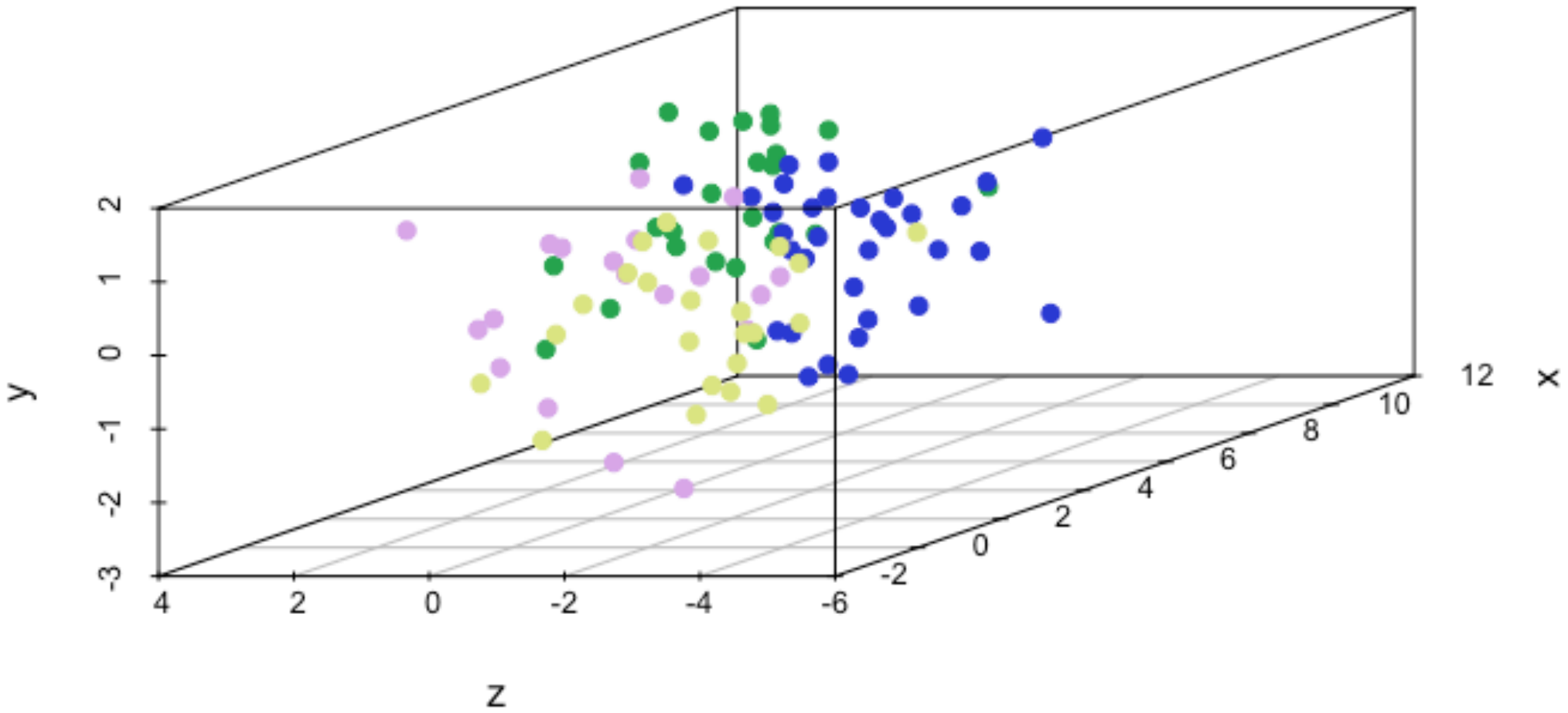
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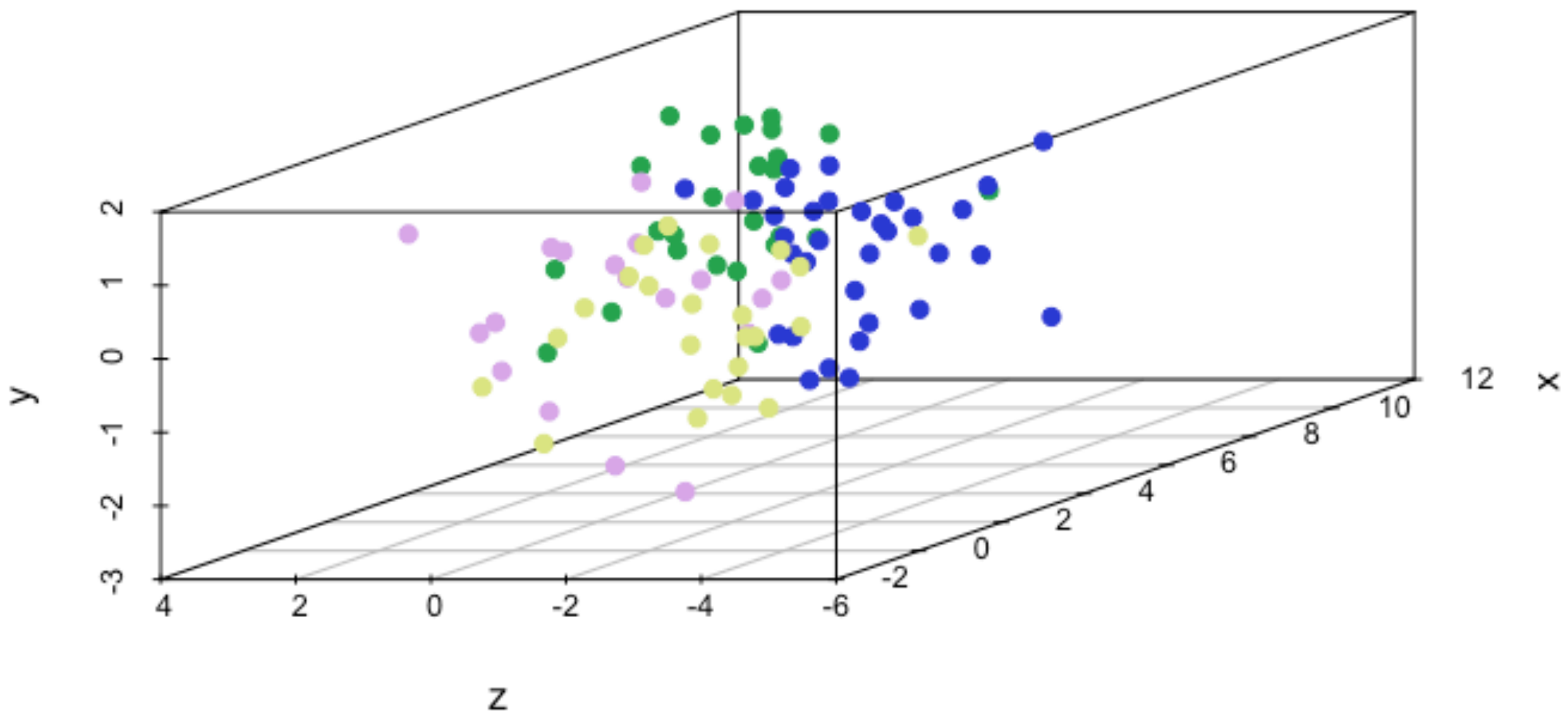
- How to cluster data when only a subset of features are relevant
- Want to identify key features and group data based on them
- Benefits of sparse clustering :
  - more accurate clusters by using only truly relevant covariates
  - results are interpretable
  - future data requires fewer features to cluster

Clustering

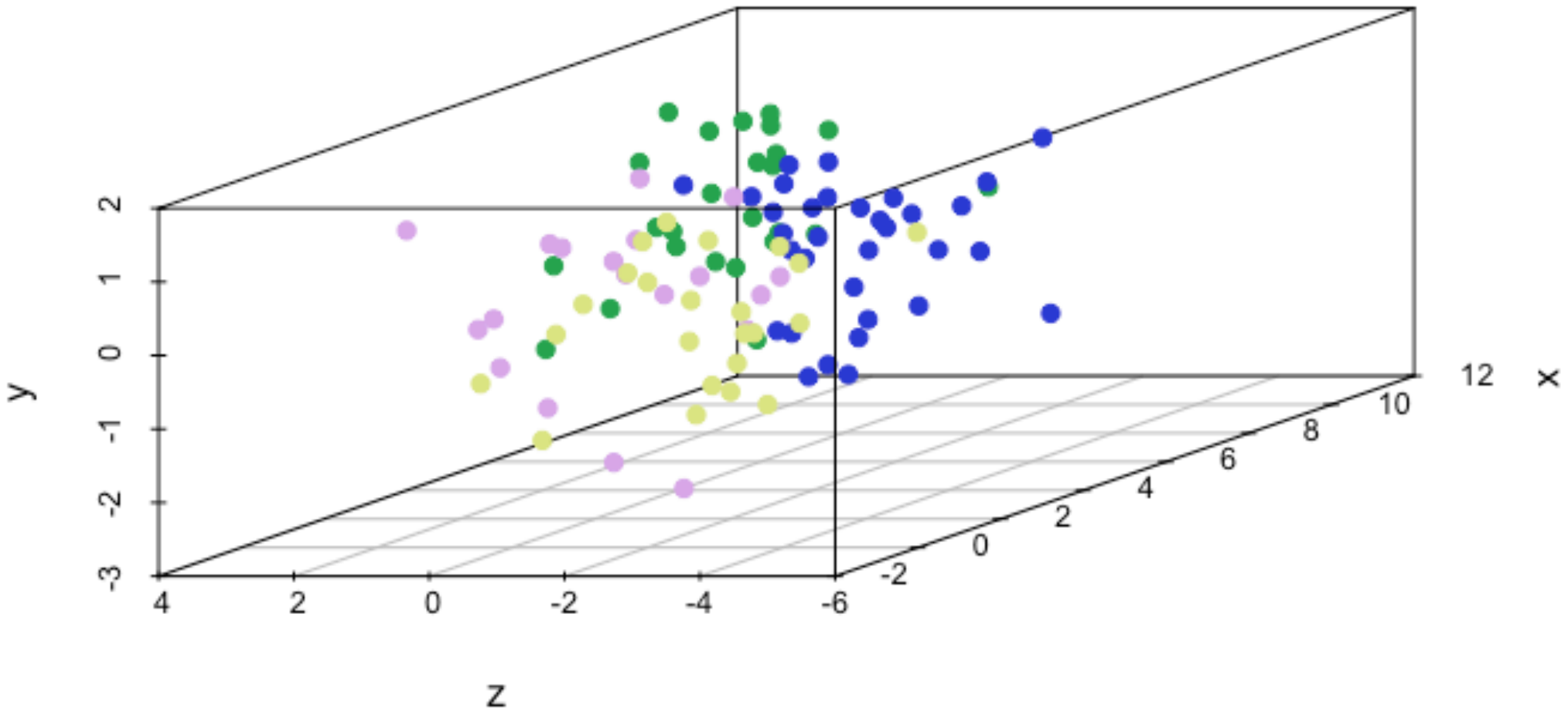


Clustering

$$d(\mathbf{x}_i, \mathbf{x}_{i'}) = \sum_{j=1}^{p} (X_{ij} - X_{i'j})^2$$

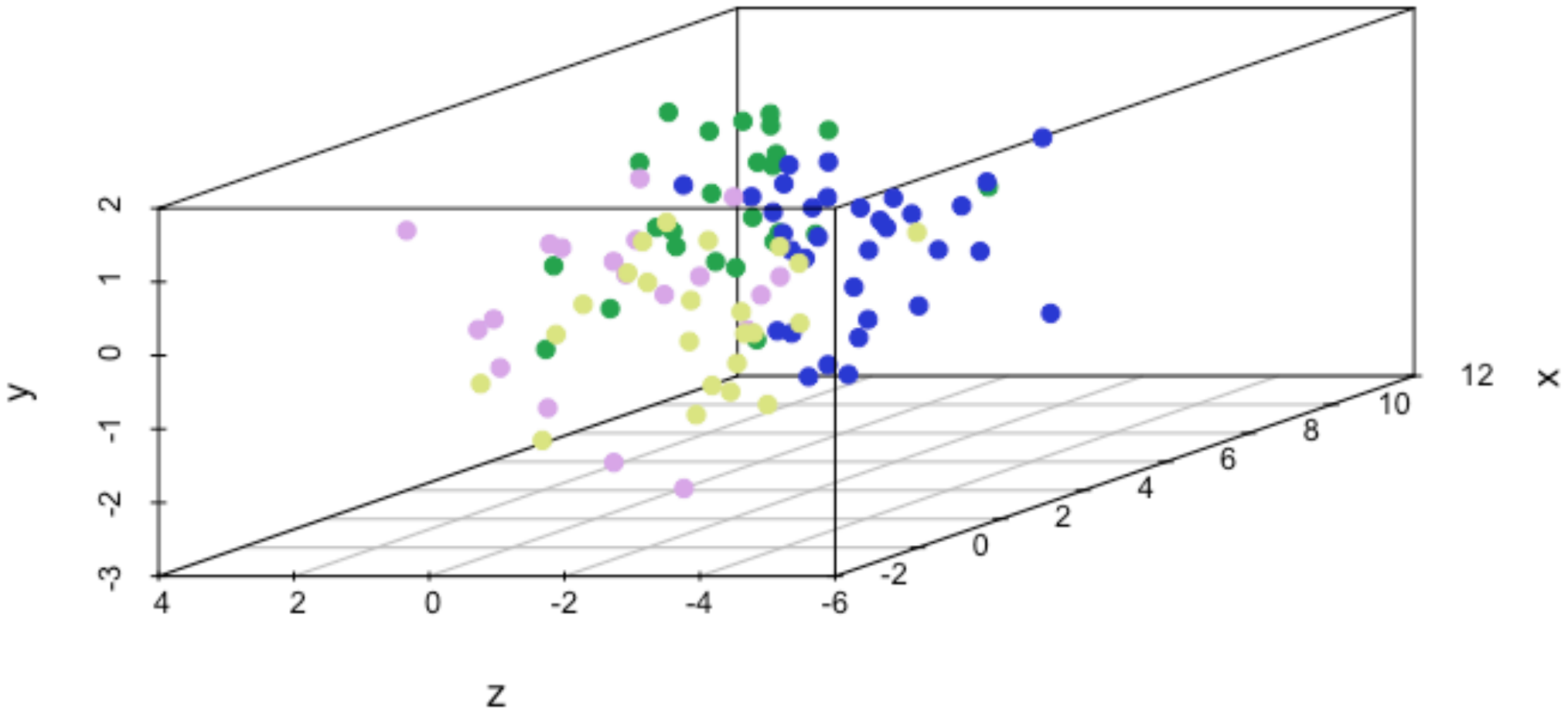


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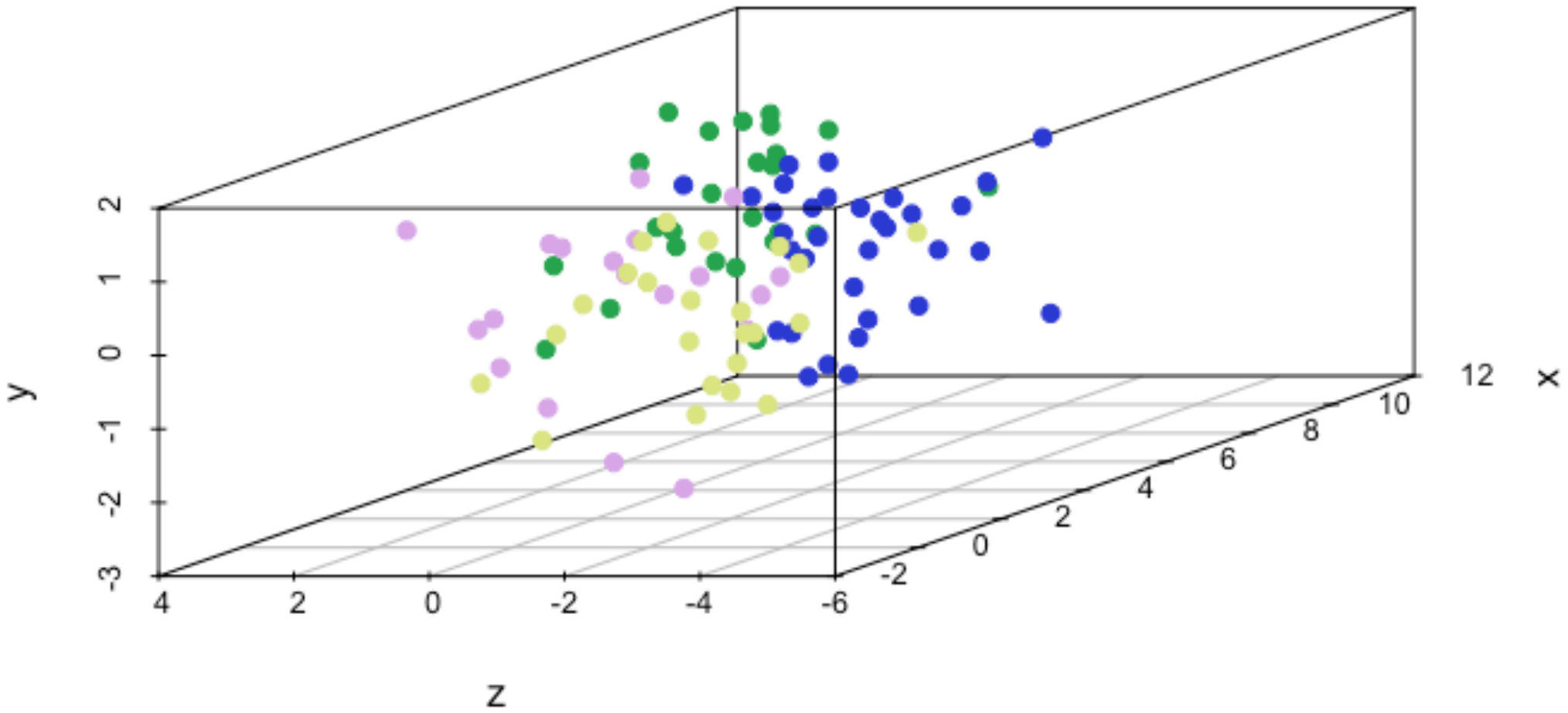


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$$d(\mathbf{x}_i, \mathbf{x}_{i'}) = \sum_{j=1}^{p} |X_{ij} - X_{i'j}|$$

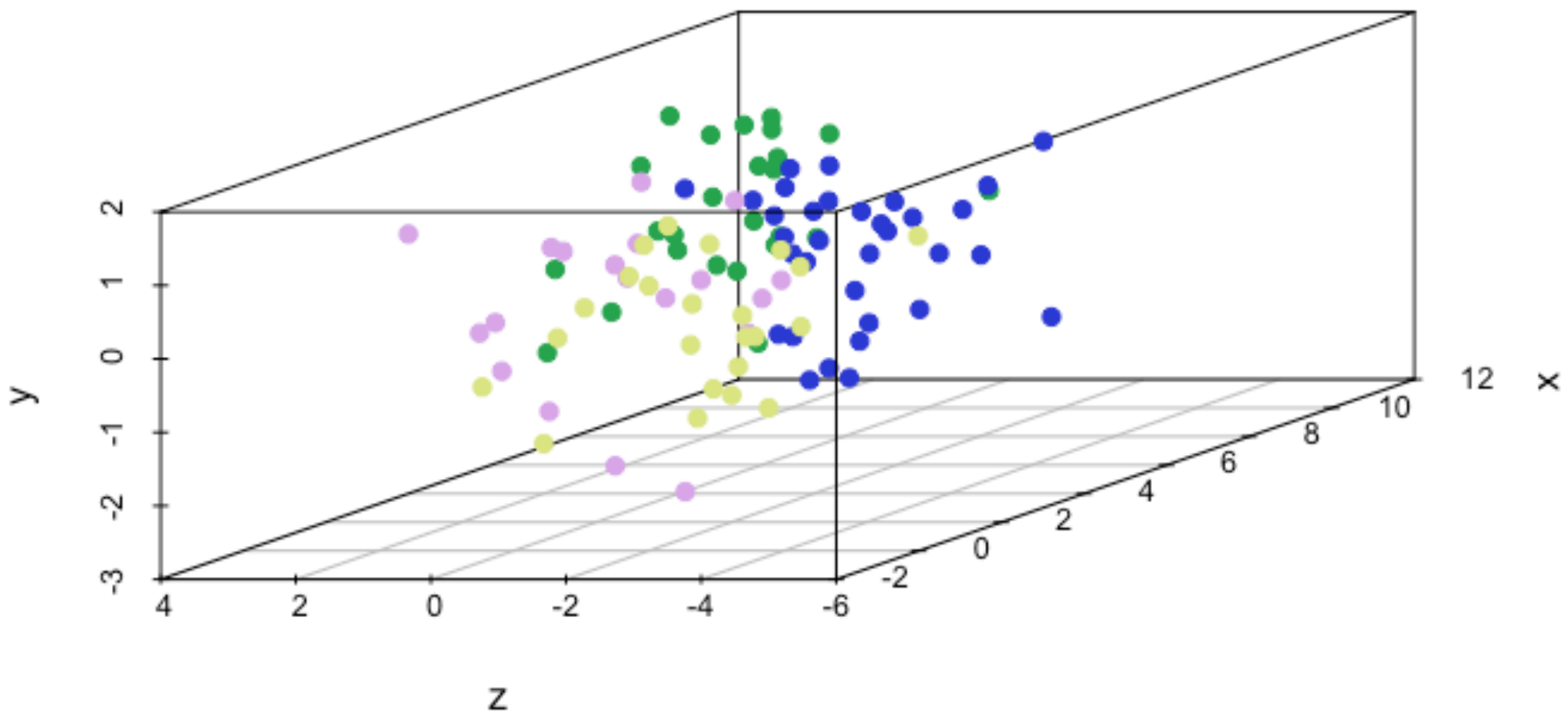


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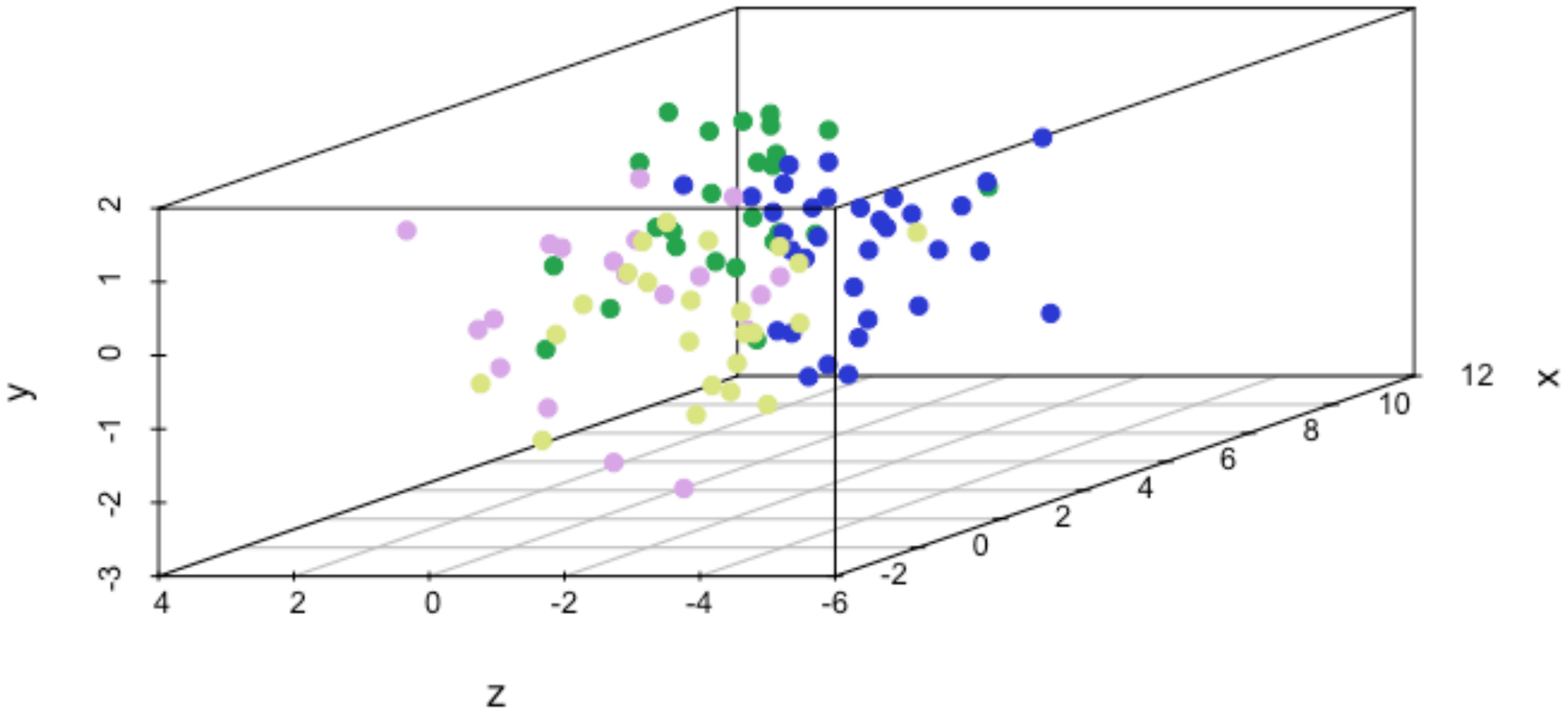


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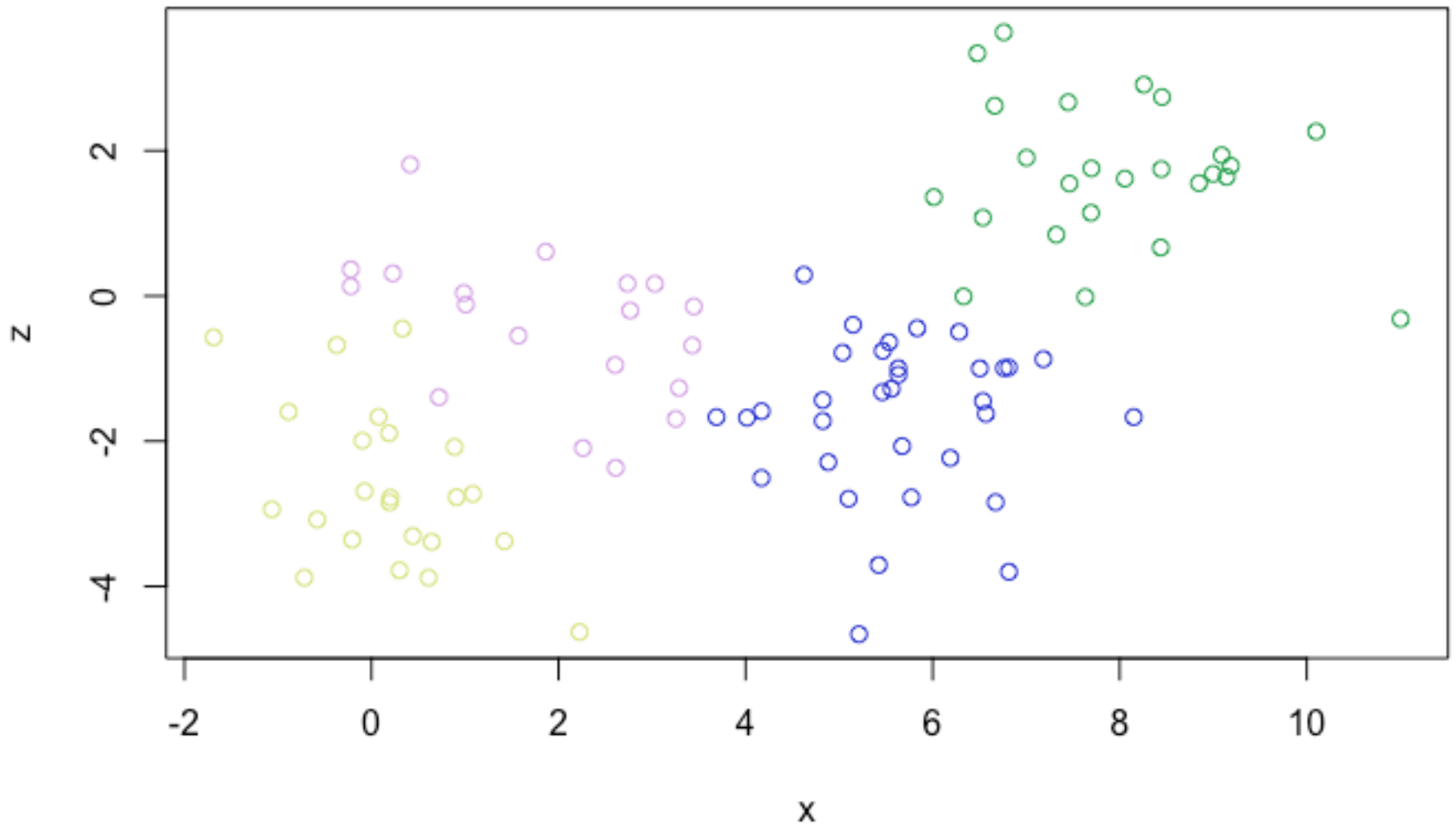
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Clustering



Clustering



Clustering

K-Means Clustering

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$$\max_{\Theta \in D} \sum_{j=1}^{p} f_j(X_j, \Theta)$$

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Within-cluster sum of squares

Clustering

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$$\min_{C} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i,i' \in C_k} \| X_i - X_{i'} \|^2$$

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BCSS = TSS - WCSS

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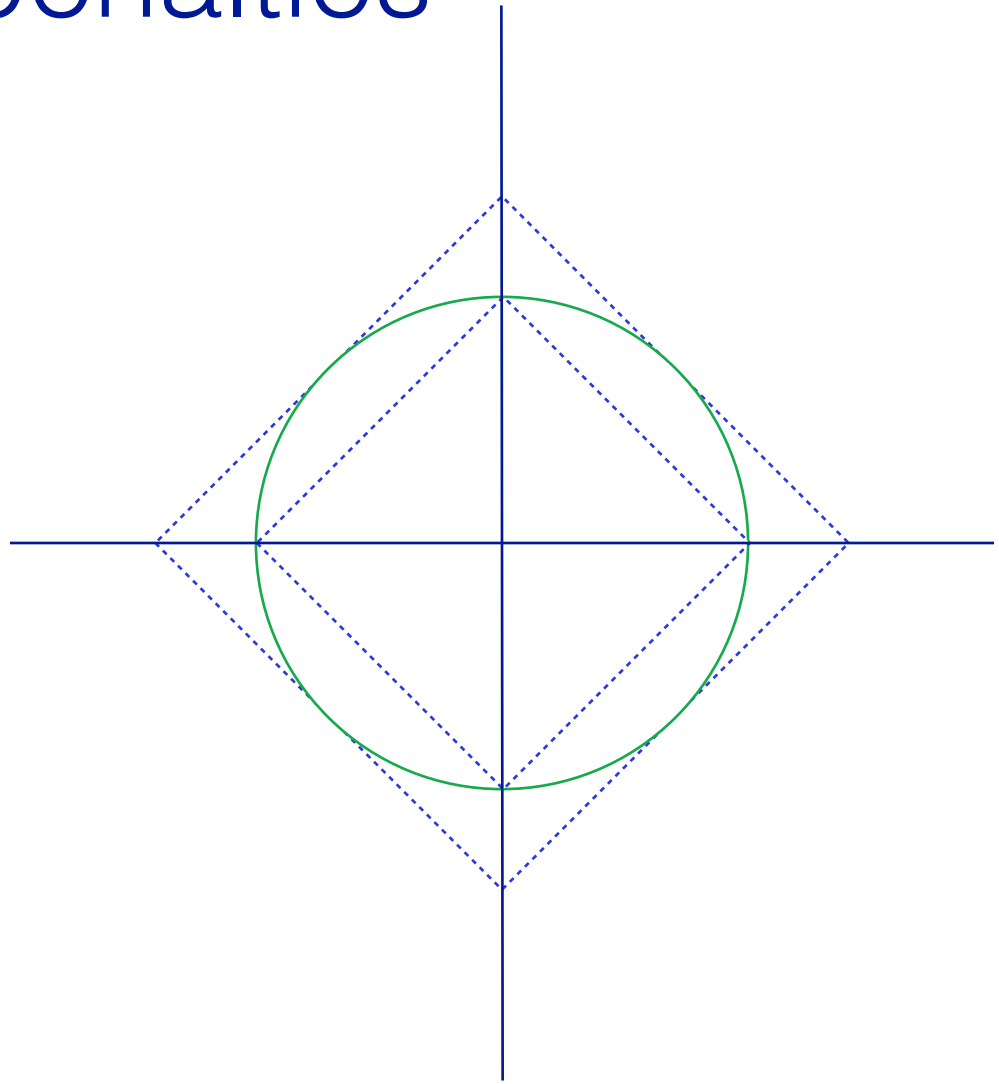
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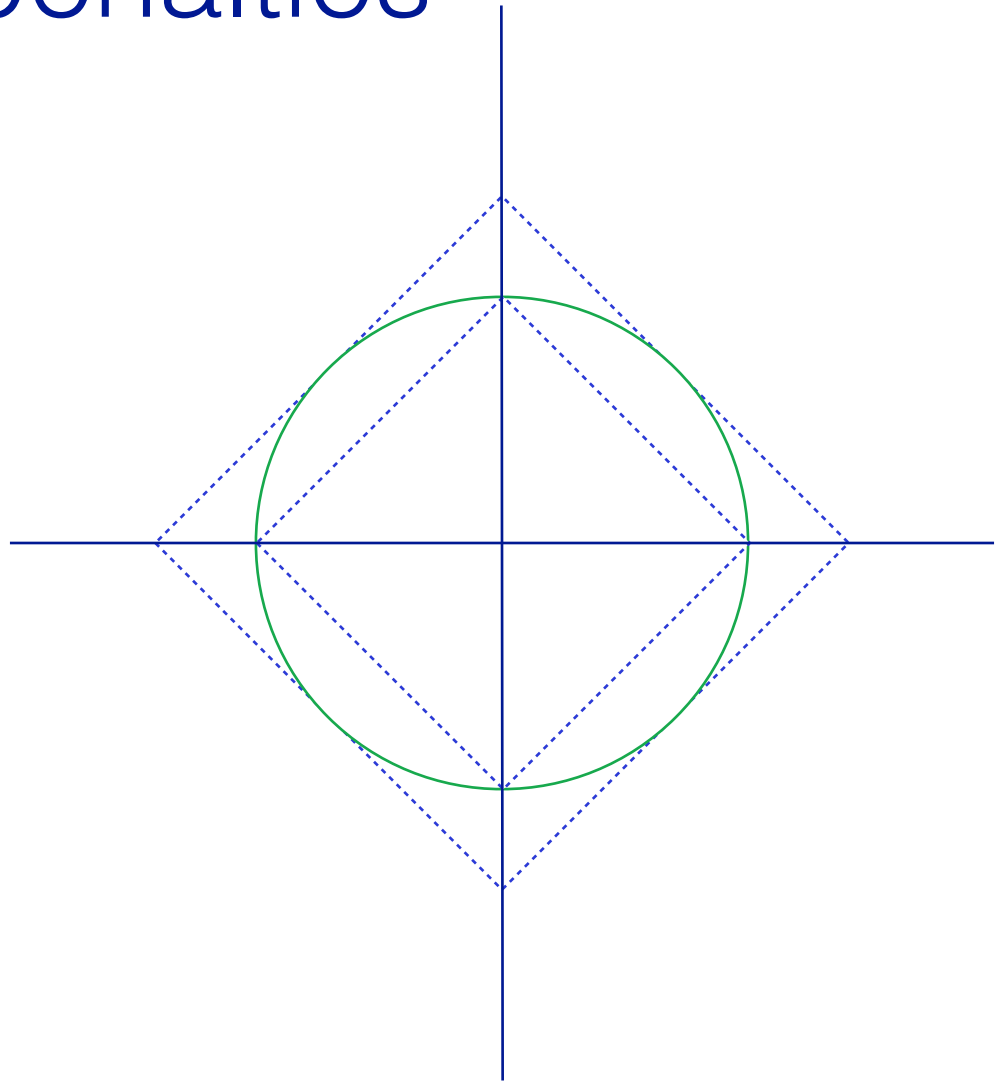
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L1 and L2 penalties

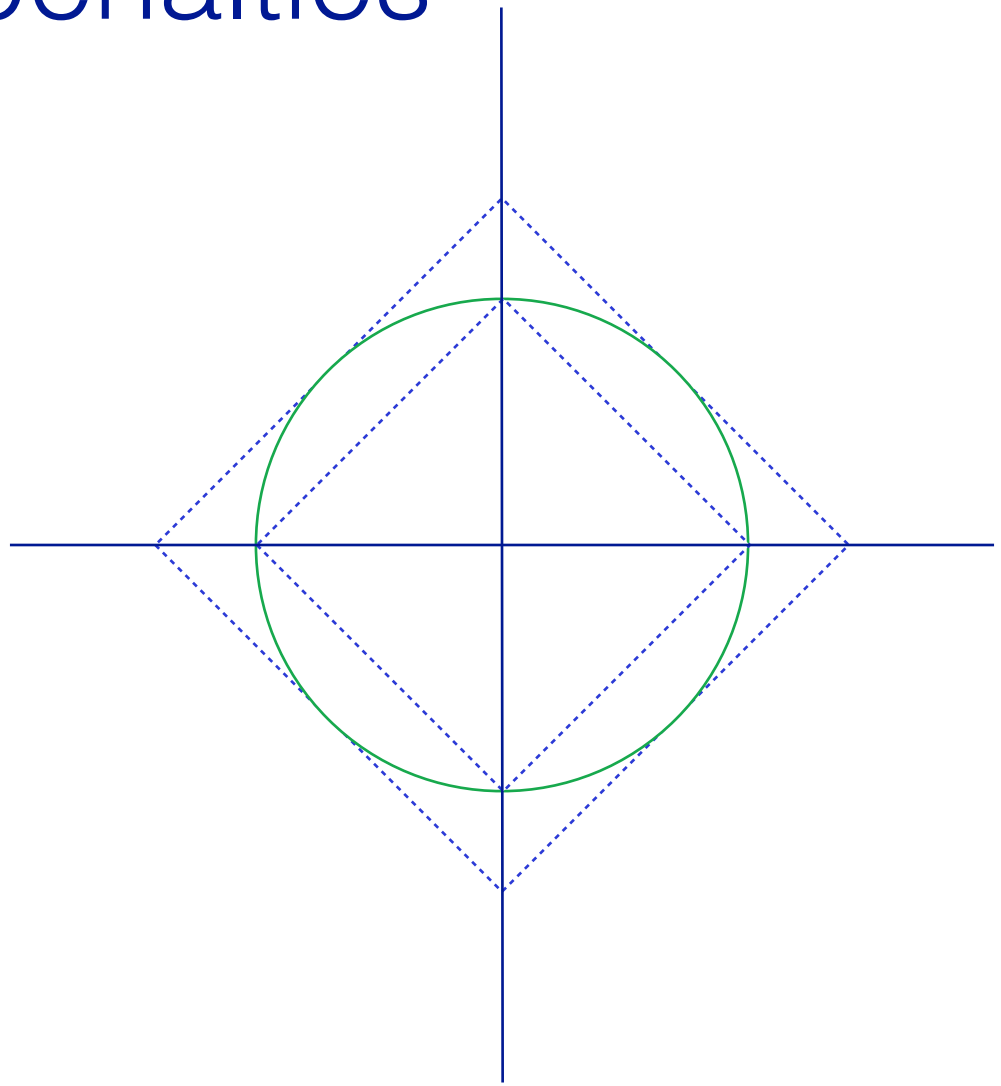


L1 and L2 penalties



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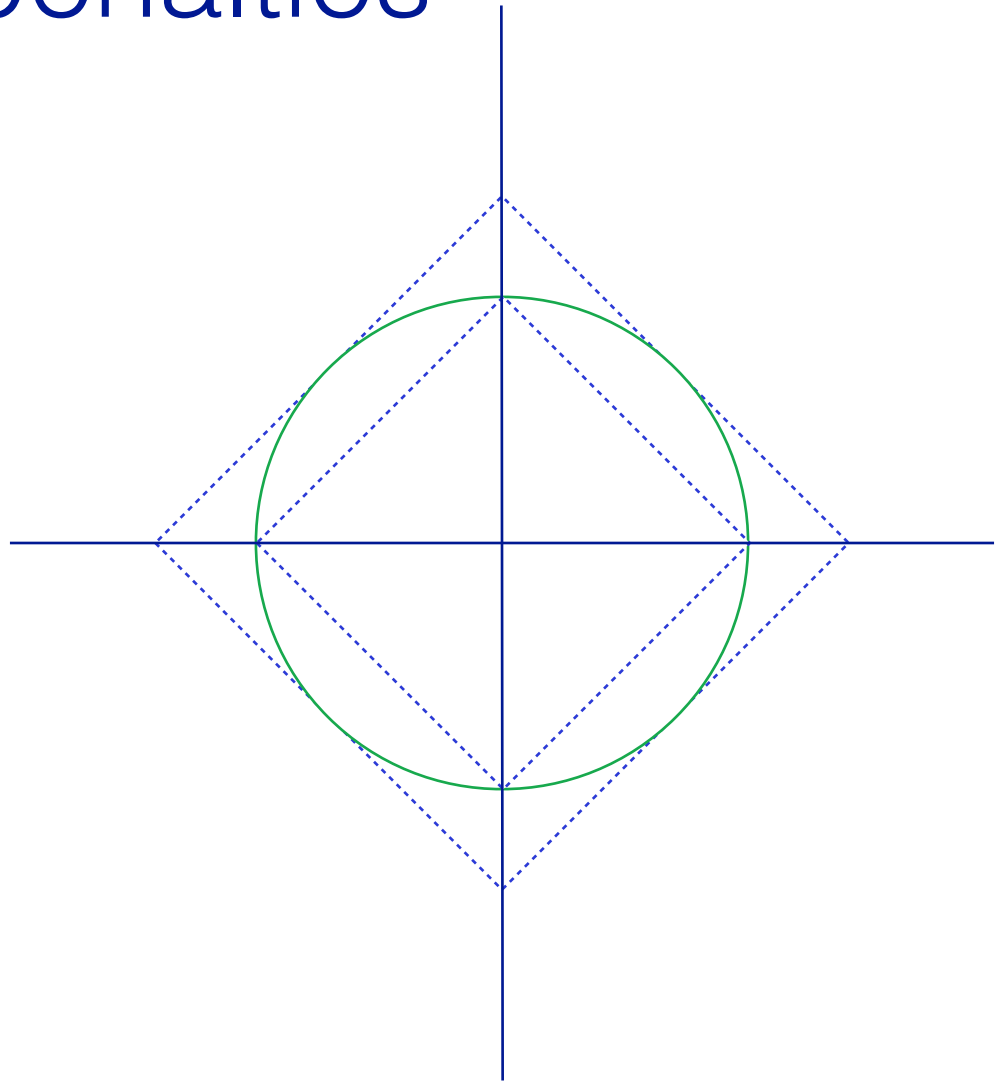
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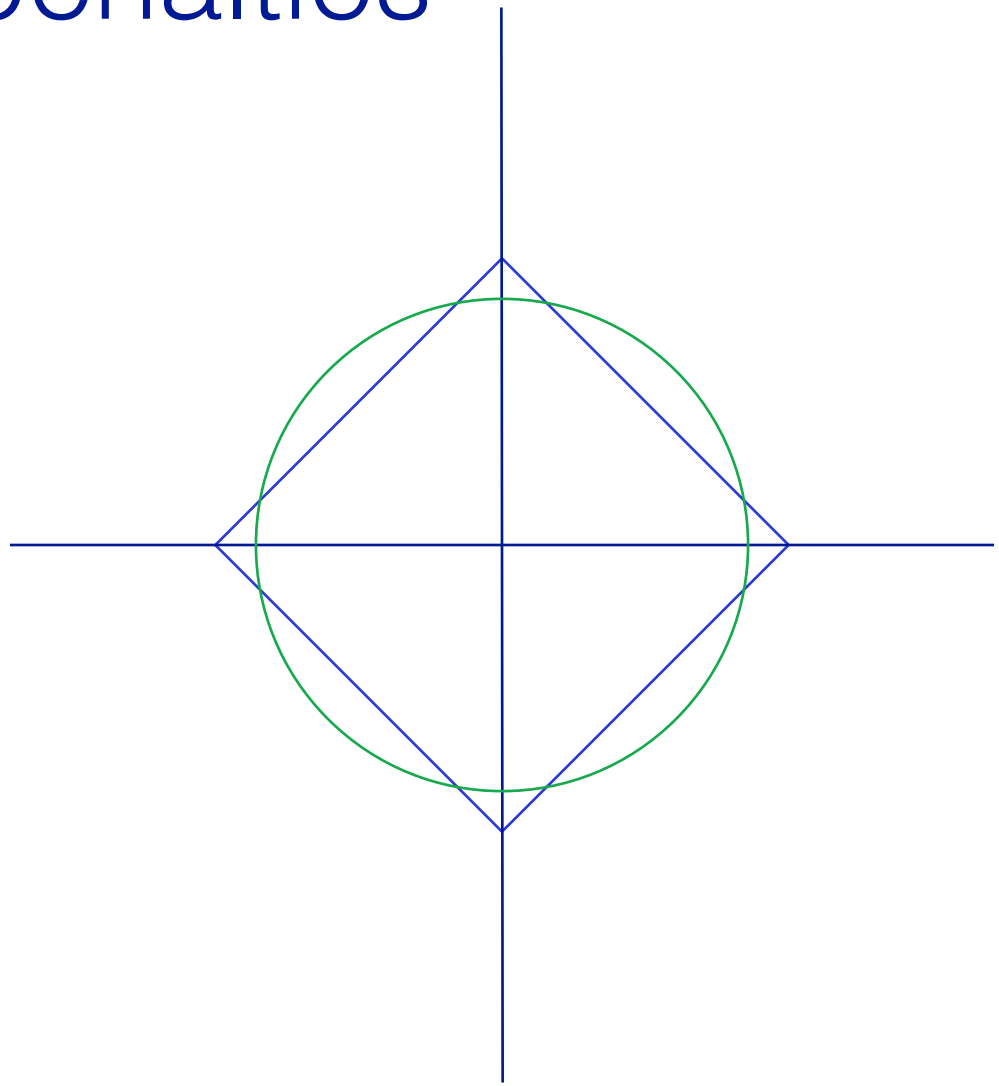


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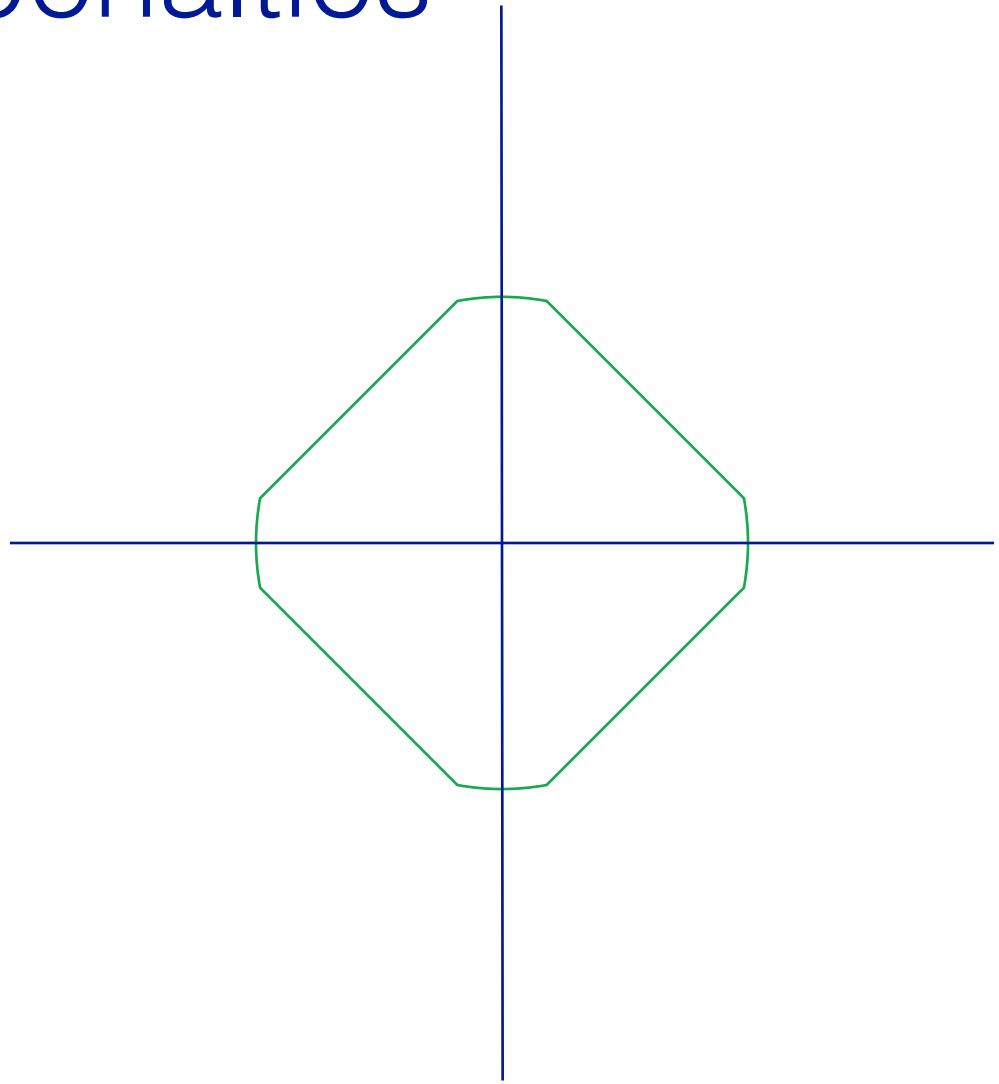
$$\| w \|_1 \leq s$$

$$1 \leq s \leq \sqrt{p}$$

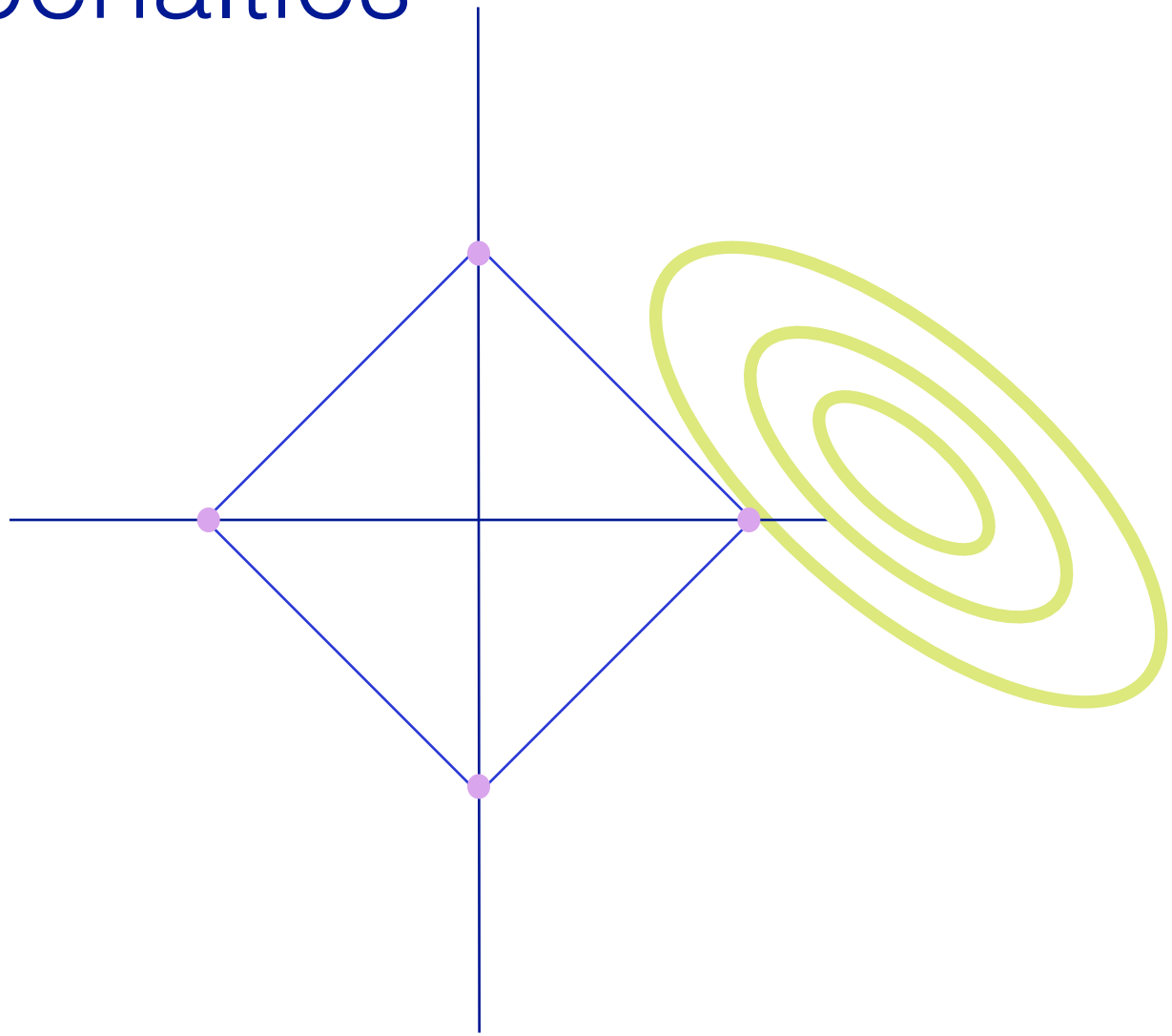
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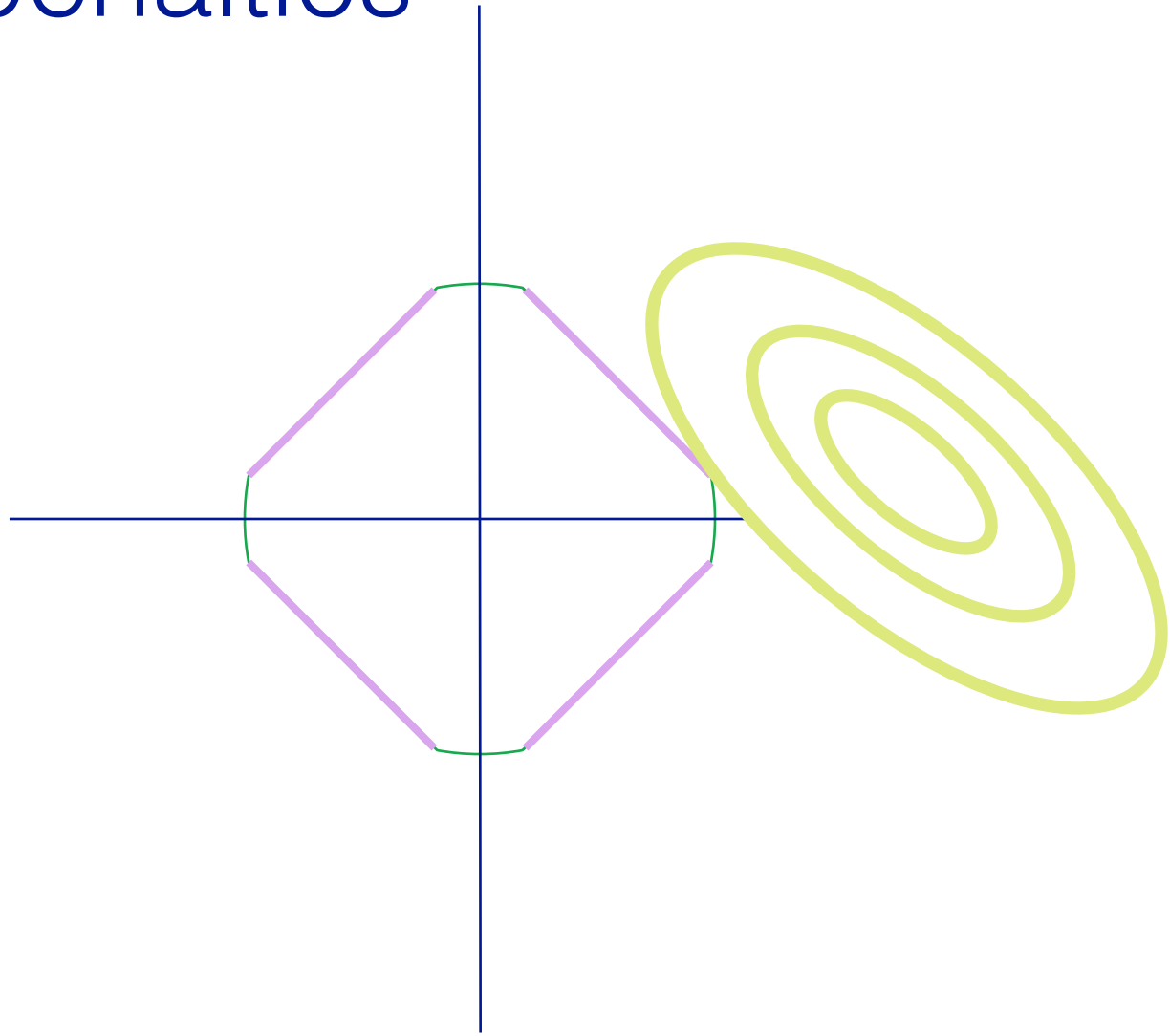
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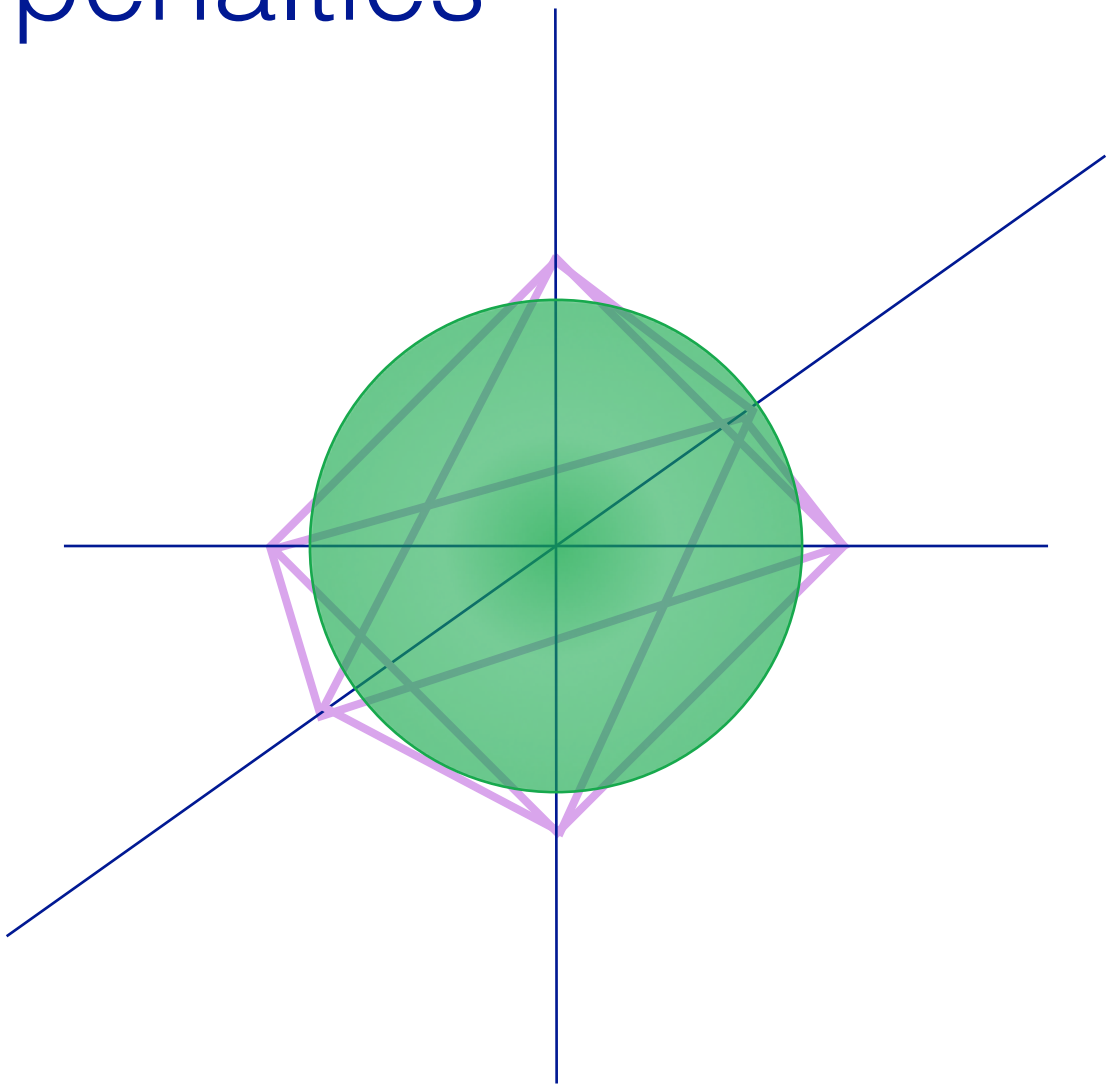
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3. Clusters given by $C_1, \ldots, C_K$, and feature weights by $w_1, \ldots, w_p$.

Hierarchical Clustering

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$$\max_{U} \sum_{j} \sum_{i,i'} d_{i,i',j} U_{i,i'}$$

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$$\max_{w,u} u^T D w \qquad \| u \|^2 \leq 1$$

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   b. For $b = 1,2,\ldots B,$ compute the objective $O_b(s)$ with sparse k-means on X_b.

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Feasible with large p			
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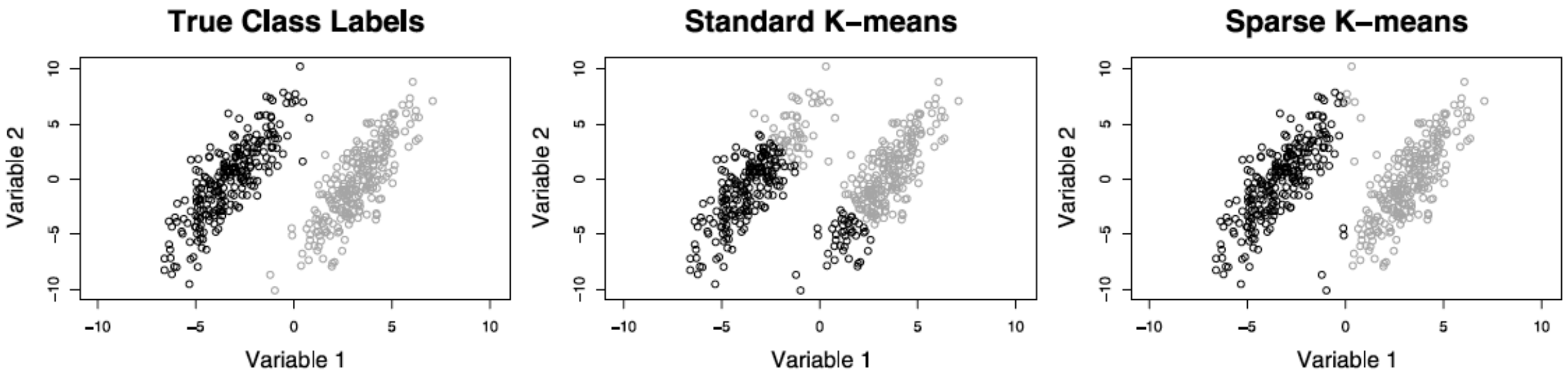
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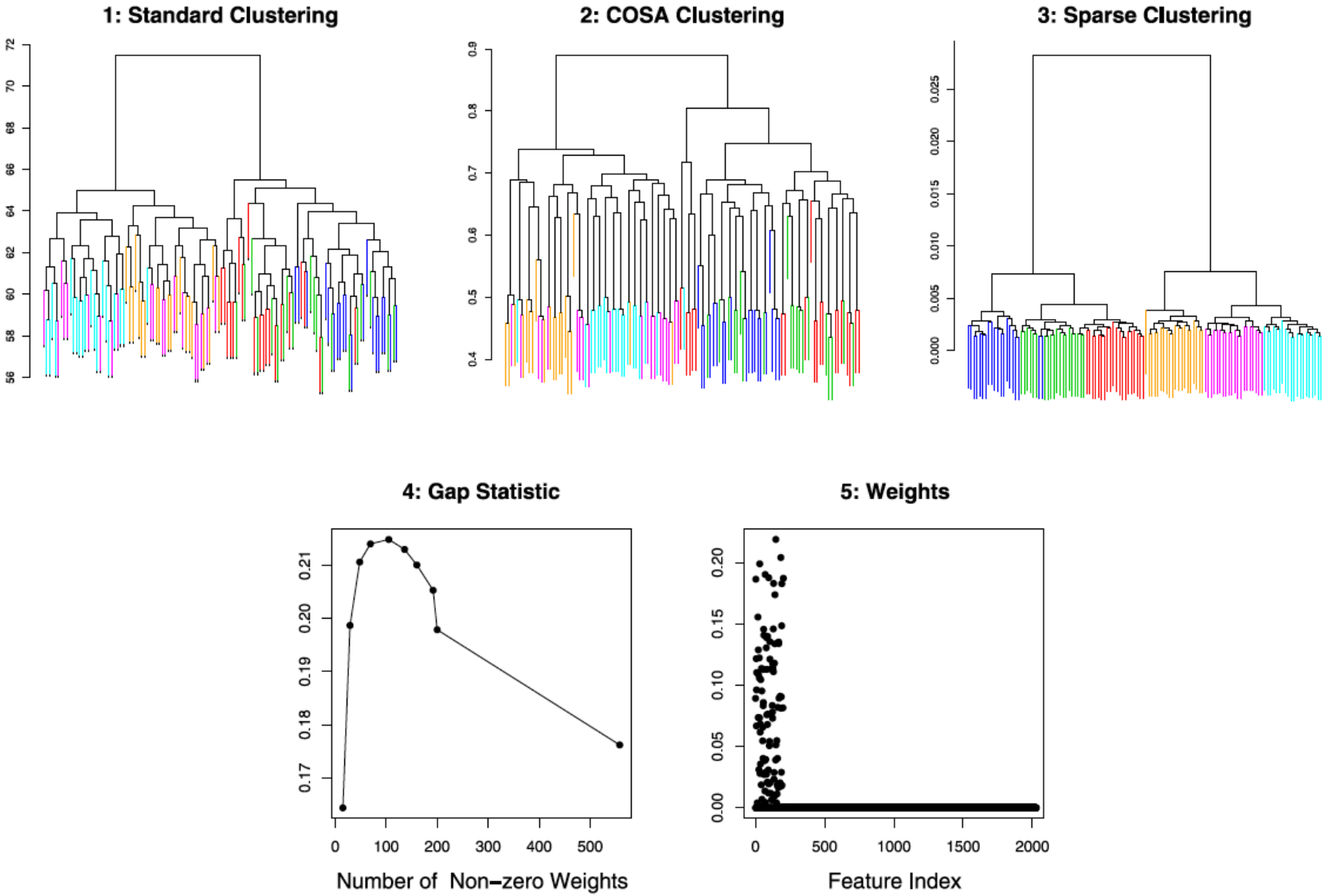
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Simulations



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Experiments

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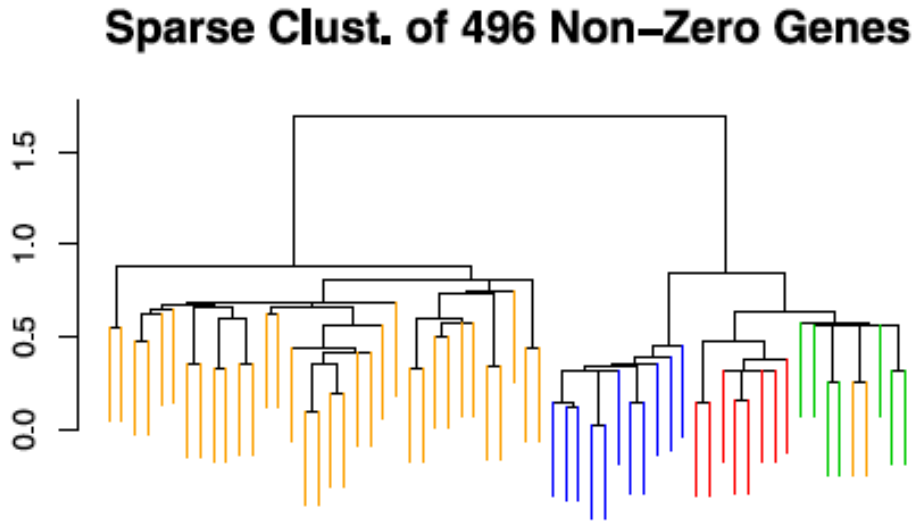
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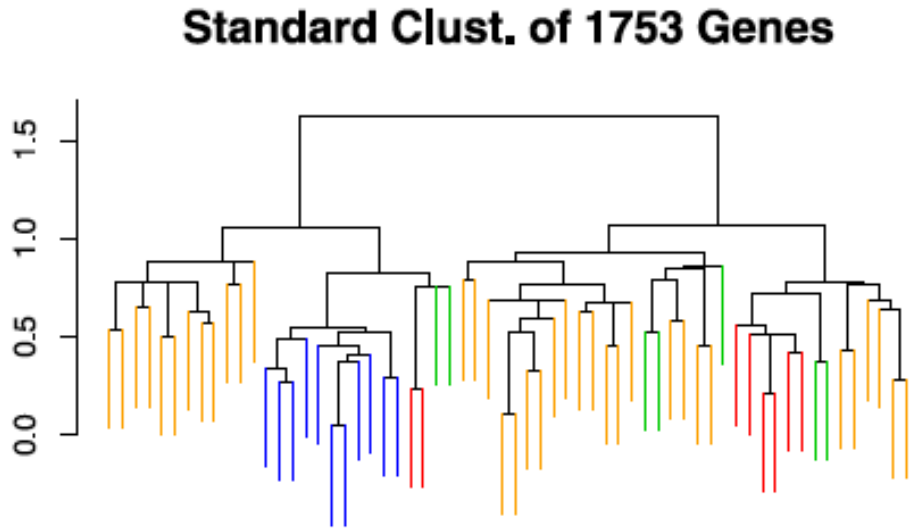
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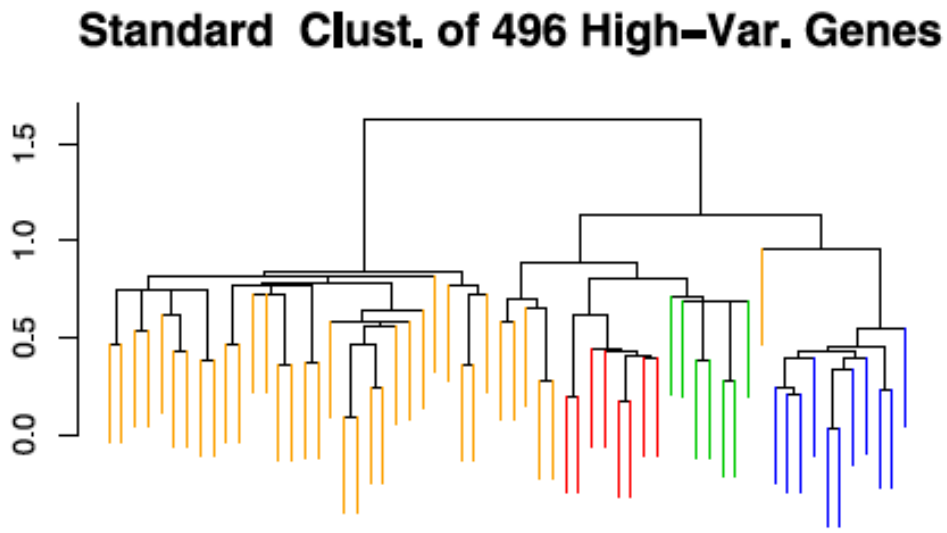
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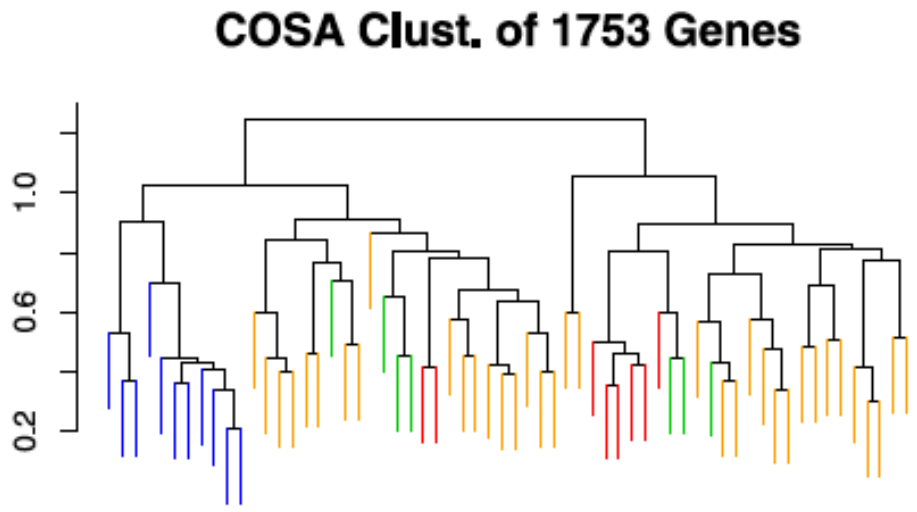
Compare standard, sparse, and COSA hierarchical clustering

Experiments

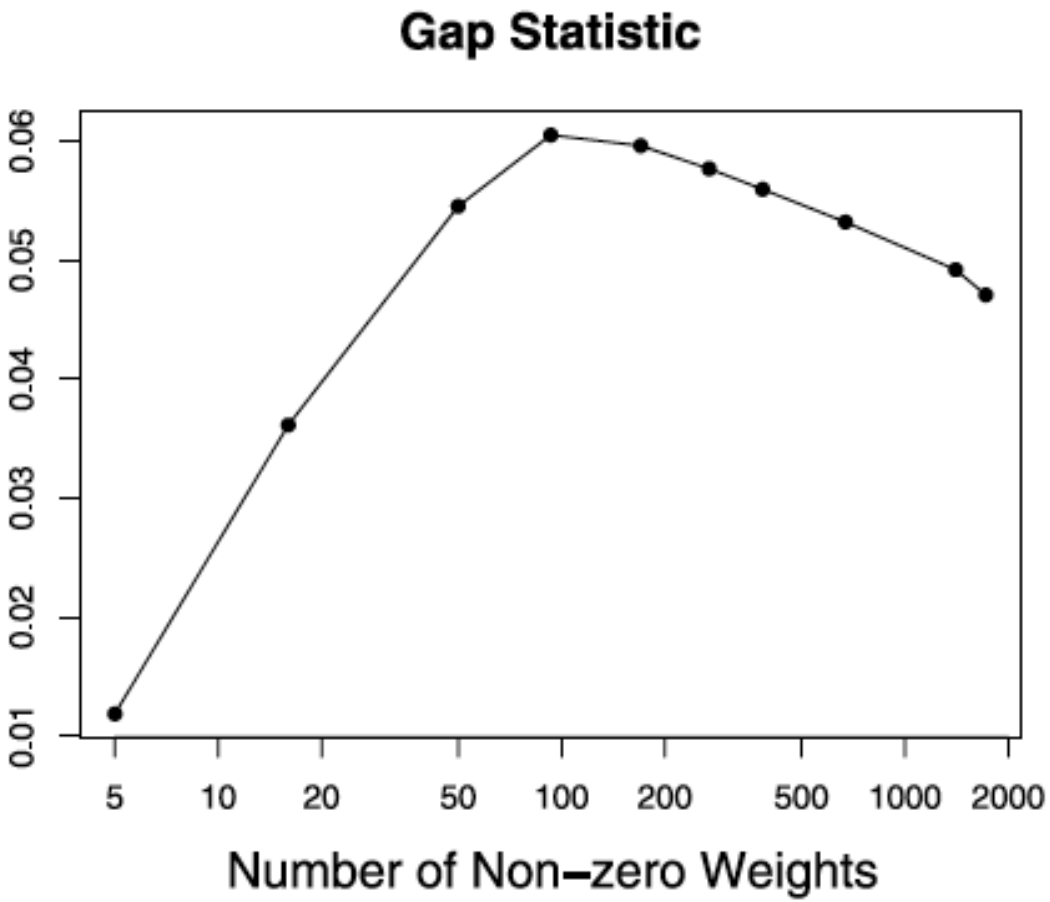








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DNA sequences belonging to 3 populations - African ancestry in southwest USA, Utah residents with European ancestry, and Han Chinese from Beijing

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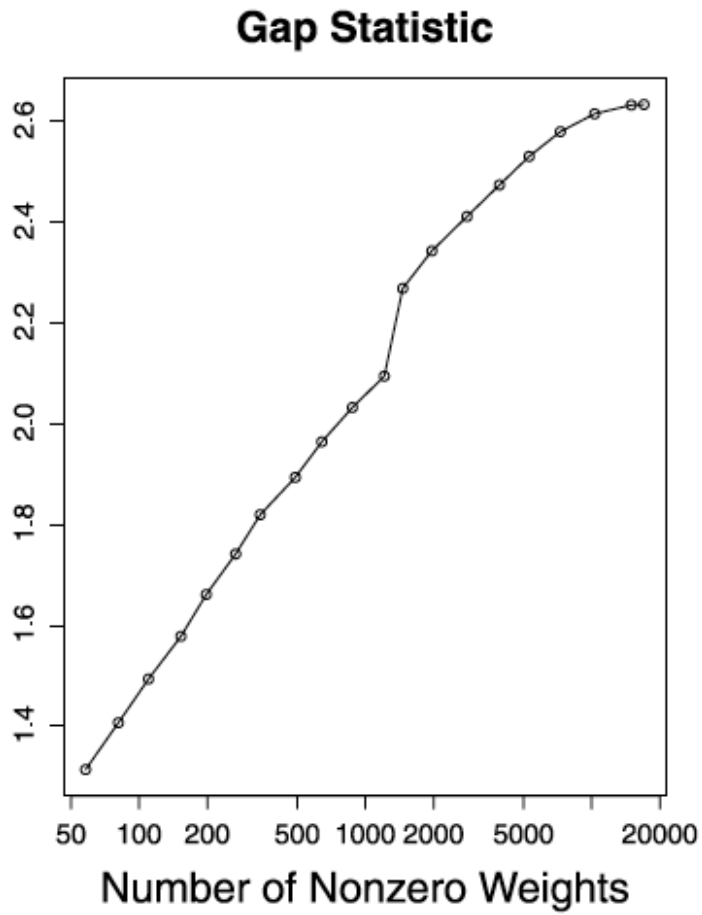
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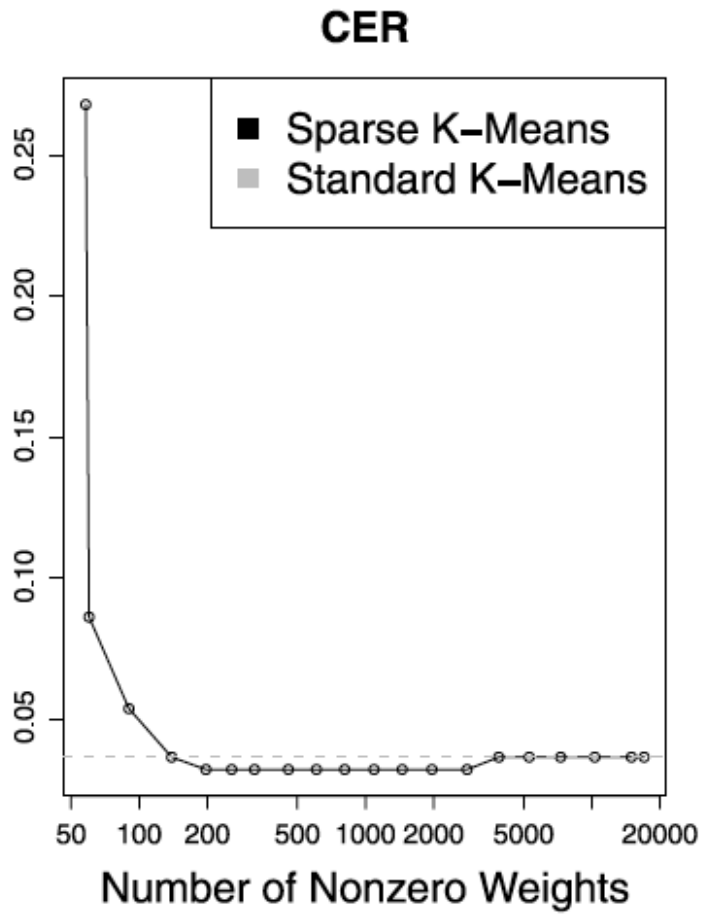
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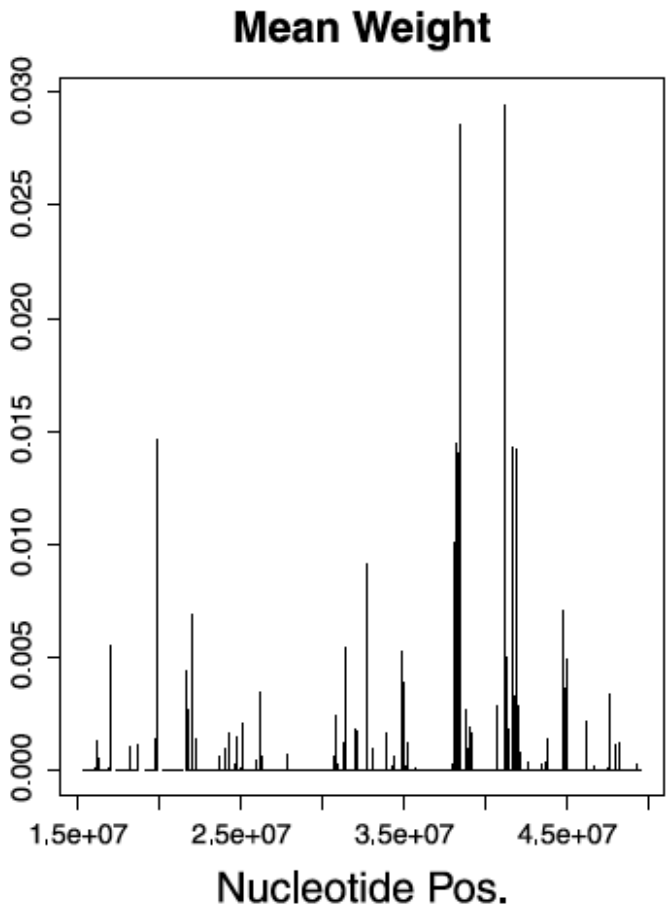
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Questions

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